# 9.6 Permutations and Combinations

Algebra I 9.6 · Student sheet

Name _____ Period ____ Date _____

**I can…**

- count the ways to arrange objects using factorials.
- find permutations and combinations and tell which one to use.
- use permutations and combinations to find probabilities.

## Part 1 · Officers vs. committee

Your group is A, B, C, and D.

1. List every way to choose a **president and a vice president** (write president first).

2. List every way to choose a **2-person committee**.

How many in each list? _____ and _____. Why is one list twice as long?

## Part 2 · Notes: fill in the blanks

n! = _____ 0! = _____

Order matters → _____: ${}_nP_r$ = _____

Order doesn't matter → _____: ${}_nC_r$ = _____

Probability = (favorable selections) ÷ _____

## Part 3 · Examples (work them with the class)

### Example 1

a. In how many ways can you arrange the letters in MATH?

b. In how many ways can you arrange 2 of the letters?

### Example 2

Ten runners are in a race. In how many ways can the gold, silver, and bronze medals be awarded?

### Example 3

A pizza shop has 8 toppings. How many different 3-topping pizzas can you order?

### Example 4

From a club of 12 members, how many ways can you

a. choose a 4-member committee?

b. choose a captain and a co-captain?

### Example 5

A class has 5 boys and 7 girls. Three students are chosen at random to present. What is the probability that all three are girls?

## Part 4 · Partner practice

### Partner 1

a. In how many ways can 6 books be arranged on a shelf?

b. In how many ways can 3 of the 6 books be arranged?

### Partner 2

A pizza shop offers 10 toppings. How many different 3-topping pizzas are possible?